

Dr. Mohamed Gad-el-Hak

Summary of Publications

- Books: 20 books.
- Journal Articles: 143 journal articles.
- Essays in Magazines and Newspapers: 53 essays.
- Book Chapters: 63 publications.
- Proceedings Articles: 73 publications.
- Refereed Meeting Articles and Abstracts: 150 articles.
- Reports: 120 reports.
- Book Reviews: 28 reviews.
- Patents: 2 granted; 7 disclosed.
- Invited Lectures: 313 lectures.

List of Publications

I. Books

1. Gad-el-Hak, M. (editor) (1989) *Advances in Fluid Mechanics Measurements, Lecture Notes in Engineering*, vol. 45, 606 pages, Springer-Verlag, New York, New York.
2. Gad-el-Hak, M. (editor) (1989) *Frontiers in Experimental Fluid Mechanics, Lecture Notes in Engineering*, vol. 46, 532 pages, Springer-Verlag, New York, New York.
3. Bonnet, J.-P., Gad-el-Hak, M., and Pollard, A. (editors) (1996) *Proceedings of the Workshop on Flow Control: Fundamentals and Practices*, 200 pages, Institut d'Etudes Scientifiques des Cargèse, Corsica, France.
4. Breuer, K.S., Bandyopadhyay, P.R., and Gad-el-Hak, M. (editors) (1996) *Application of Microfabrication to Fluid Mechanics*, DSC-Volume 59, 468 pages, ASME, New York, New York.
5. Gad-el-Hak, M., Pollard, A., and Bonnet, J.-P. (editors) (1998) *Flow Control: Fundamentals and Practices, Lecture Notes in Physics*, vol. m53, 540 pages, Springer-Verlag, Berlin.
6. Gad-el-Hak, M. (guest editor) (1998) *Flow Control: Fundamentals and Practices*, special issue of *Experimental Thermal and Fluid Science (ETFS)*, vol. 16, no. 1&2, 172 pages, Elsevier, New York, New York.
7. Gad-el-Hak, M. (2000) *Flow Control: Passive, Active, and Reactive Flow Management*, 448 pages, Cambridge University Press, London, United Kingdom. Reprinted in paperback 2006.
8. Gad-el-Hak, M. (editor) (2002) *The MEMS Handbook*, 1368 pages, CRC Press, Boca Raton, Florida. Second edition 2006.
9. Gad-el-Hak, M. (editor) (2004) *Proceedings of the International Conference on Transport Phenomena in Micro- and Nanodevices*, CD Publication, Engineering Conference International, Brooklyn, New York.

10. Gad-el-Hak, M. (guest editor) (2005) *Special Topics: Transport Phenomena in Micro- and Nanodevices*, special issue of *Physics of Fluids*, vol. 17, no. 10, 139 pages, American Institute of Physics, New York, New York.
11. Gad-el-Hak, M. (editor) (2006) *MEMS: Introduction and Fundamentals*, 448 pages, CRC Taylor & Francis, Boca Raton, Florida.
12. Gad-el-Hak, M. (editor) (2006) *MEMS: Design and Fabrication*, 664 pages, CRC Taylor & Francis, Boca Raton, Florida.
13. Gad-el-Hak, M. (editor) (2006) *MEMS: Applications*, 568 pages, CRC Taylor & Francis, Boca Raton, Florida.
14. Gad-el-Hak, M., and Tsai, H.M. (editors) (2006) *Transition and Turbulence Control*, 444 pages, World Scientific, Singapore.
15. Gad-el-Hak, M. (editor) (2006) *Proceedings of the U.S.–Egypt Workshop on Predictive Methodologies for Global Weather-Related Disasters*, CD Publication, Virginia Commonwealth University, Richmond, Virginia.
16. Gad-el-Hak, M. (editor) (2006) *Proceedings of the Second International Conference on Transport Phenomena in Micro- and Nanodevices*, CD Publication, Engineering Conference International, Brooklyn, New York.
17. Gad-el-Hak, M. (guest editor) (2007) *Transport Phenomena in Micro- and Nanodevices*, special issue of *Nanoscale and Microscale Thermophysical Engineering*, vol. 11, no. 1–2, 226 pages, Taylor & Francis, New York, New York.
18. Gad-el-Hak, M. (editor) (2008) *Large-Scale Disasters: Prediction, Control, and Mitigation*, 600 pages, Cambridge University Press, London, United Kingdom.
19. Kizilova, N., and Gad-el-Hak, M. (editors) (2012) *Contemporary Problems of Mathematics, Mechanics and Computing Sciences*, Kharkov University Press, Kharkov, Ukraine.
20. Castillo, L., Carbajal, G., and Gad-el-Hak, M. (editors) (2015) *TTU & WindInspire Summer Research Institute Proceedings: Renewable Energy & medicine*, Tex Tech University, Lubbock, Texas.

II. Journal Articles

1. Gad-el-Hak, M., and Corrsin, S. (1974) “Measurements of the Nearly Isotropic Turbulence Behind a Uniform Jet Grid,” *Journal of Fluid Mechanics* **62**, pp. 115–143.
2. Kutchai, H., Morton, J.B., and Gad-el-Hak, M. (1977) “Turbulent-Flow Properties of Dilute Solutions of Red Blood-Cells,” *Biophysical Journal* **17**, p. A260.
3. Gad-el-Hak, M., Morton, J.B., and Kutchai, H. (1977) “Turbulent-Flow of Red-Cells in Dilute Suspensions: Effect on Kinetics of O₂ Uptake,” *Biophysical Journal* **18**, pp. 289–300.
4. Howard, A.D., Morton, J.B., Gad-el-Hak, M., and Pierce, D.B. (1978) “Sand Transport Model of Barchan Dune Equilibrium,” *Sedimentology* **25**, pp. 307–338.
5. Gad-el-Hak, M., and Morton, J.B. (1979) “Experiments on the Diffusion of Smoke in Isotropic Turbulent Flow,” *AIAA Journal* **17**, pp. 558–562.
6. Gad-el-Hak, M., Blackwelder, R.F., and Riley, J.J. (1981) “On the Growth of Turbulent Regions in Laminar Boundary Layers,” *Journal of Fluid Mechanics* **110**, pp. 73–95.
7. Gad-el-Hak, M., Davis, S.H., McMurray, J.T., and Orszag, S.A. (1984) “On the Stability of the Decelerating Boundary Layer,” *Journal of Fluid Mechanics* **138**, pp. 297–323.

8. Lin, J.-T., and Gad-el-Hak, M. (1984) "Turbulence Characteristics in Wind-Waves," *Journal of Geophysical Research* **89**, no. C1, pp. 627–636.
9. Gad-el-Hak, M., Blackwelder, R.F., and Riley, J.J. (1984) "On the Interaction of Compliant Coatings with Boundary Layer Flows," *Journal of Fluid Mechanics* **140**, pp. 257–280.
10. Gad-el-Hak, M., and Blackwelder, R.F. (1985) "The Discrete Vortices from a Delta Wing," *AIAA Journal* **23**, pp. 961–962.
11. Gad-el-Hak, M., and Ho, C.-M. (1985) "The Pitching Delta Wing," *AIAA Journal* **23**, pp. 1660–1665.
12. Gad-el-Hak, M. (1986) "The Response of Elastic and Viscoelastic Surfaces to a Turbulent Boundary Layer," *Journal of Applied Mechanics* **53**, pp. 206–212.
13. Gad-el-Hak, M. (1986) "The Use of the Dye-Layer Technique for Unsteady Flow Visualization," *Journal of Fluids Engineering* **108**, pp. 34–38.
14. Gad-el-Hak, M. (1986) "Boundary Layer Interactions with Compliant Coatings: An Overview," feature article in *Applied Mechanics Reviews* **39**, pp. 511–524.
15. Gad-el-Hak, M., and Ho, C.-M. (1986) "Unsteady Vortical Flow Around Three-Dimensional Lifting Surfaces," *AIAA Journal* **24**, pp. 713–721.
16. Gad-el-Hak, M., and Ho, C.-M. (1986) "Unsteady Flow Around an Ogive-Cylinder," *Journal of Aircraft* **23**, pp. 520–528.
17. Gad-el-Hak, M., and Hussain, A.K.M.F. (1986) "Coherent Structures in a Turbulent Boundary Layer. Part 1: Generation of 'Artificial' Bursts," *Physics of Fluids* **29**, pp. 2124–2139.
18. Gad-el-Hak, M. (1987) "Compliant Coatings Research: A Guide to the Experimentalist," *Journal of Fluids and Structures* **1**, pp. 55–70.
19. Gad-el-Hak, M. (1987) "Unsteady Separation on Lifting Surfaces," feature article in *Applied Mechanics Reviews* **40**, pp. 441–453.
20. Gad-el-Hak, M. (1987) "The Water Towing Tank as an Experimental Facility: An Overview," *Experiments in Fluids* **5**, pp. 289–297.
21. Gad-el-Hak, M., and Blackwelder, R.F. (1987) "Control of the Discrete Vortices from a Delta Wing," *AIAA Journal* **25**, pp. 1042–1049.
22. Gad-el-Hak, M., and Blackwelder, R.F. (1987) "Simulation of Large-Eddy Structures in a Turbulent Boundary Layer," *AIAA Journal* **25**, pp. 1207–1215.
23. Riley, J.J., Gad-el-Hak, M., and Metcalfe, R.W. (1988) "Compliant Coatings," *Annual Review of Fluid Mechanics* **20**, pp. 393–420.
24. Gad-el-Hak, M. (1988) "Visualization Techniques for Unsteady Flows: An Overview," *Journal of Fluids Engineering* **110**, pp. 231–243.
25. Gad-el-Hak, M., and Blackwelder, R.F. (1989) "Selective Suction for Controlling Bursting Events in a Boundary Layer," *AIAA Journal* **27**, pp. 308–314.
26. Gad-el-Hak, M. (1989) "Flow Control," feature article in *Applied Mechanics Reviews* **42**, pp. 261–293.
27. Gad-el-Hak, M. (1990) "Large Gradients due to Wall Turbulence," in *Some Unanswered Questions in Fluid Mechanics*, eds. L.M. Trefethen and R.L. Panton, *Applied Mechanics Reviews* **43**, pp. 153–170.
28. Gad-el-Hak, M. (1990) "Control of Low-Speed Airfoil Aerodynamics," *AIAA Journal* **28**, pp. 1537–1552.

29. Xu, J.C., Sen, M., and Gad-el-Hak, M. (1990) "Low-Reynolds Number Flow Over a Rotatable Cylinder-Splitter Plate Body," *Physics of Fluids A* **2**, pp. 1925–1927.
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33. Gad-el-Hak, M., and Bandyopadhyay, P.R. (1994) "Questions in Fluid Mechanics: Reynolds Number Effects in Wall-Bounded Flows," *Journal of Fluids Engineering* **116**, pp. 2–3.
34. Zaki, T.G., Sen, M., and Gad-el-Hak, M. (1994) "Numerical and Experimental Investigation of Flow Past a Freely Rotatable Square Cylinder," *Journal of Fluids and Structures* **8**, pp. 555–582.
35. Gad-el-Hak, M., and Bandyopadhyay, P.R. (1994) "Reynolds Number Effects in Wall-Bounded Flows," feature article in *Applied Mechanics Reviews* **47**, pp. 307–365.
36. Gad-el-Hak, M. (1994) "Interactive Control of Turbulent Boundary Layers: A Futuristic Overview," *AIAA Journal* **32**, pp. 1753–1765.
37. Gad-el-Hak, M., and Bandyopadhyay, P.R. (1995) "Field Versus Laboratory Turbulent Boundary Layers," *AIAA Journal* **33**, pp. 361–364.
38. Gad-el-Hak, M. (1995) "Questions in Fluid Mechanics: Stokes' Hypothesis for a Newtonian, Isotropic Fluid," *Journal of Fluids Engineering* **117**, pp. 3–5.
39. Gad-el-Hak, M., and Sen, M. (1996) "Fluid Mechanics in the Next Century," *Applied Mechanics Reviews* **49**, no. 3, pp. III–IV.
40. Bandyopadhyay, P.R., and Gad-el-Hak, M. (1996) "Rotating Gas-Liquid Flows in Finite Cylinders: Sensitivity of Standing Vortices to End Effects," *Experiments in Fluids* **21**, pp. 124–138.
41. Gad-el-Hak, M. (1996) "Modern Developments in Flow Control," *Applied Mechanics Reviews* **49**, pp. 365–379.
42. Sen, M., Wajerski, D., and Gad-el-Hak, M. (1996) "A Novel Pump for MEMS Applications," *Journal of Fluids Engineering* **118**, pp. 624–627.
43. Gad-el-Hak, M. (1996) "Compliant Coatings: A Decade of Progress," *Applied Mechanics Reviews* **49**, no. 10, part 2, pp. S147–S157.
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45. Sharatchandra, M.C., Sen, M., and Gad-el-Hak, M. (1997) "Navier-Stokes Simulations of a Novel Viscous Pump," *Journal of Fluids Engineering* **119**, pp. 372–382.
46. Maureau, J., Sharatchandra, M.C., Sen, M., and Gad-el-Hak, M. (1997) "Flow and Load Characteristics of Microbearings with Slip," *Journal of Micromechanics and Microengineering* **7**, pp. 55–64.
47. Gad-el-Hak, M., and Leissa, A.W. (1997) "Introduction to George Batchelor's retrospective 'Research as a Life Style'," *Applied Mechanics Reviews* **50**, no. 8, pp. R11–R20.
48. Gad-el-Hak, M. (1997) "The Last Conundrum," *Applied Mechanics Reviews* **50**, no. 12, part 1, pp. 1–2.
49. Sharatchandra, M.C., Sen, M., and Gad-el-Hak, M. (1998) "New Approach to Constrained Shape Optimization Using Genetic Algorithms," *AIAA Journal* **36**, pp. 51–61.

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55. Gad-el-Hak, M. (1998) "Fluid Mechanics from the Beginning to the Third Millennium," *International Journal of Engineering Education* **14**, pp. 177–185.
56. Löfdahl, L., and Gad-el-Hak, M. (1999) "MEMS Applications in Turbulence and Flow Control," *Progress in Aerospace Sciences* **35**, pp. 101–203.
57. Gad-el-Hak, M. (1999) "The Fluid Mechanics of Microdevices—The Freeman Scholar Lecture," *Journal of Fluids Engineering* **121**, pp. 5–33.
58. Löfdahl, L., and Gad-el-Hak, M. (1999) "MEMS-Based Pressure and Shear Stress Sensors," *Measurement Science and Technology* **10**, pp. 665–686.
59. Gad-el-Hak, M. (2001) "Flow Control: The Future," *Journal of Aircraft* **38**, pp. 402–418.
60. Gad-el-Hak, M. (2001) "Micro-Air-Vehicles: Can They be Controlled Better?," *Journal of Aircraft* **38**, pp. 419–429.
61. Gad-el-Hak, M. (2001) "Physique des Écoulements dans les MEMS," *Mécanique et Industries* **2**, pp. 313–341.
62. Gad-el-Hak, M. (2002) "Compliant Coatings for Drag Reduction," *Progress in Aerospace Sciences* **38**, pp. 77–99.
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65. Buschmann, M.H., and Gad-el-Hak, M. (2003) "Generalized Logarithmic Law and Its Consequences," *AIAA Journal* **41**, pp. 40–48.
66. Buschmann, M.H., and Gad-el-Hak, M. (2003) "Debate Concerning the Mean-Velocity Profile of a Turbulent Boundary Layer," *AIAA Journal* **41**, pp. 565–572.
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68. Hamadiche, M., and Gad-el-Hak, M. (2004) "Spatiotemporal Stability of Flow Through Collapsible, Viscoelastic Tubes," *AIAA Journal* **42**, pp. 772–786.
69. Gad-el-Hak, M. (2004) "Transport Phenomena in Microdevices," *Zeitschrift für Angewandte Mathematik und Mechanik (ZAMM)* **84**, pp. 494–498.

70. Buschmann, M.H., and Gad-el-Hak, M. (2004) "Comment on 'Evaluating the Law of the Wall in Two-Dimensional Fully-Developed Turbulent Channel Flows'," *Physics of Fluids* **16**, pp. 3507–3508.
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72. Buschmann, M.H., and Gad-el-Hak, M. (2005) "New Mixing-Length Approach for the Mean Velocity Profile of Turbulent Boundary Layers," *Journal of Fluids Engineering* **127**, pp. 393–396.
73. Gad-el-Hak, M. (2005) "Preface: Transport Phenomena in Micro- and Nanodevices," *Physics of Fluids* **17**, p. 100501.
74. Gad-el-Hak, M. (2005) "Liquids: The Holy Grail of Microfluidic Modeling," *Physics of Fluids* **17**, pp. 100612.1–100612.13.
75. Gad-el-Hak, M. (2005) "Differences Between Liquid and Gas Transport at the Microscale," *Bulletin of the Polish Academy of Sciences* **53**, pp. 301–316.
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78. Indinger, T., Buschmann, M.H., and Gad-el-Hak, M. (2006) "Mean-Velocity Profile of Turbulent Boundary Layers Approaching Separation," *AIAA Journal* **44**, pp. 2465–2474.
79. Buschmann, M.H., and Gad-el-Hak, M. (2006) "Structure of the Canonical Turbulent Wall-Bounded Flow," *AIAA Journal* **44**, pp. 2500–2503.
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81. Buschmann, M.H., and Gad-el-Hak, M. (2007) "Scaling of the Mean-Velocity Profiles of the Canonical Turbulent Wall-Bounded Flow," *Progress in Aerospace Sciences* **42**, pp. 419–467.
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84. Buschmann, M.H., and Gad-el-Hak, M. (2009) "Evidence of Non-Logarithmic Behavior of Turbulent Channel and Pipe Flow," *AIAA Journal* **47**, pp. 535–541.
85. Gad-el-Hak, M. (2009) "The Art and Science of Large-Scale Disasters," *Bulletin of the Polish Academy of Sciences* **57**, pp. 3–34.
86. Hamadiche, M., Kizilova, N., and Gad-el-Hak, M. (2009) "Suppression of Absolute Instabilities in the Flow Inside a Compliant Tube," *Communications in Numerical Methods in Engineering* **25**, pp. 505–531.
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94. Buschmann, M.H., and Gad-el-Hak, M. (2010) "Kolmogorov Scaling of Turbulent Flow in the Vicinity of the Wall," *Physica D: Nonlinear Phenomena* **239**, pp. 1288–1295.
95. Buschmann, M.H., and Gad-el-Hak, M. (2010) "Normal and Cross-Flow Reynolds Stresses: Differences Between Confined and Semi-Confined Flows," *Experiments in Fluids* **49**, pp. 213–223.
96. Bhadauria, R., Pidaparti, R.M., and Gad-el-Hak, M. (2010) "Solution of Two-Dimensional Viscous Flow Driven by Motion of Flexible Walls," *CFD Letters* **2**, pp. 1–12.
97. Trimble, S.W., Grody, W.W., McKelvey, B., and Gad-el-Hak, M. (2010) "The Glut of Academic Publishing: A Call for a New Culture," *Academic Questions* **23**, pp. 276–286.
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101. Samaha, M.A., Ochanda, F.O., Vahedi Tafreshi, H., Tepper, G.C., and Gad-el-Hak, M. (2011) "In Situ, Non-invasive Characterization of Superhydrophobic Coatings," *Review of Scientific Instruments* **82**, pp. 045109.1–045109.7.
102. Gad-el-Hak, M. (2011) "Large-Scale Disasters: Mechanistic Framework for Prediction, Control and Mitigation," *Journal of Critical Incident Analysis* **1**, pp. 105–160.
103. Emami, B., Vahedi Tafreshi, H., Gad-el-Hak, M., and Tepper, G.C. (2011) "Predicting Shape and Stability of Air–Water Interface on Superhydrophobic Surfaces With Randomly Distributed, Dissimilar Posts," *Applied Physics Letters* **98**, pp. 203106.1–203106.3.
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